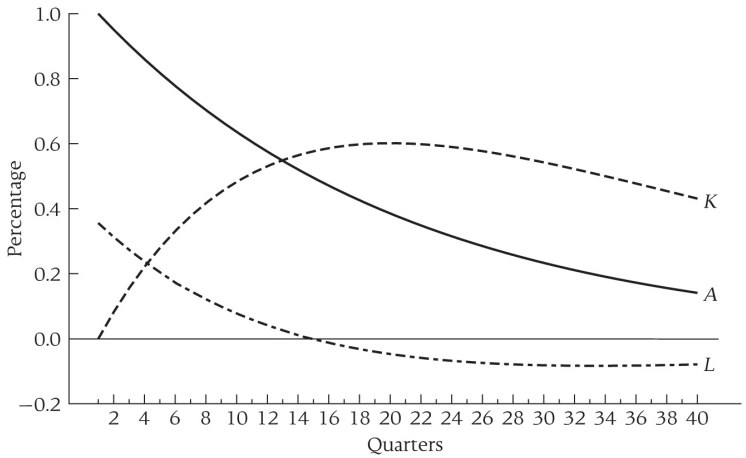


Advanced Macroeconomics

Section 4 - Fluctuations (I): RBC theory

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Influential stylized facts

1. Significant short-run fluctuations ('cycles') around growth path, but no apparent regularity
   - inspired models of an economy hit by exogenous shocks
   - shocks then propagate through endogenous mechanics
   - New-Classical vs New-Keynesian models: different assumptions on the nature of these shocks and their propagation mechanism

Influential stylized facts

1. Significant short-run fluctuations ('cycles') around growth path, but no apparent regularity
   - inspired models of an economy hit by exogenous shocks
   - shocks then propagate through endogenous mechanics
   - New-Classical vs New-Keynesian models: different assumptions on the nature of these shocks and their propagation mechanism
2. Investment much more pro-cyclical than consumption
3. Productivity is pro-cyclical
4. Real wages don't react much to short-run fluctuations (slightly pro-cyclical)

New-Classical RBC

Sources of Shocks

- Technology shocks: exogenous changes in productivity.
- Other real shocks (e.g. preferences, government spending).

Propagation Mechanisms

- Intertemporal substitution in labour supply, consumption and investment.
- Perfectly competitive markets and flexible prices: output moves with productivity.

New Keynesian DSGE

Sources of Shocks

- Nominal and demand shocks: monetary policy, nominal demand, expectations.
- Cost-push or markup shocks.

Propagation Mechanisms

- Nominal rigidities: sticky prices and wages slow adjustment.
- Imperfect competition
- Monetary policy feedback.
- Expectation dynamics

RBC theory: overview

- A Ramsey (growth) model with two modifications:
  1. temporary random shocks to 'real' parameters – typically productivity (A);
  2. endogenous labour supply: people choose how much to work.
- Shocks propagate as people optimally adjust consumption, labour and investment
  - short-run fluctuations = *optimal* responses to temporary shocks;
  - policy implication: stabilization policies (monetary & fiscal) are not needed or helpful.

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- Methodologically, it won the day:
  - supplanted old-Keynesian models without micro-foundations;
  - New-Keynesian DSGE models = RBC + rigidities/imperfections.
- But (almost) no one considers it even remotely realistic:
  - No involuntary unemployment? Monetary policy can never affect any real variable?
  - Recessions as sudden decreases in technical knowledge? (more on this at the end)

The plan

1. A simple **two-period** RBC model (Kurlat, ch. 13):
   - set-up; graphical and algebraic solution; effects of shocks.

2. The full, **infinite-horizon** RBC model (Romer, ch. 5):
   - main mechanisms and results.

3. **Discussion**:
   - calibration, policy implications, extensions and criticisms.

Set-up

- Idea of RBC: take the model we used for long-run growth and use it to study the short run (competitive markets, flexible prices, no money).
- Simplest possible version: two periods, a representative household and a representative firm.
- **Period 1**: production uses only labour

$$Y_1 = F_1(L)$$

  - the economy starts with no capital but can produce with labor only

- **Period 2**: production uses only capital

$$Y_2 = F_2(K)$$

  - K = investment made in period 1.

First Welfare Theorem: with competitive markets and no externalities, the market equilibrium = what a benevolent planner would choose.

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The planner's problem

$$\max_{c_1,c_2,l,L,K} \; u(c_1) + v(l) + \beta u(c_2)$$

$$\text{s.t.} \qquad c_1 + K = Y_1, \qquad Y_1 = F_1(L), \qquad c_2 = F_2(K), \qquad L + l = 1$$

- l = leisure;
- L = labour;
- $u(c)$ = utility from consumption;
- $v(l)$ = utility from leisure;
- β = discount factor ($0 < \beta < 1$).

Solving the planner's problem: the Lagrangian

- Use the simple constraints to substitute out $Y_1 = F_1(L)$, $c_2 = F_2(K)$ and $l = 1 - L$. One constraint is left: $c_1 + K = F_1(L)$.
- Lagrangian:

$$\mathcal{L} = u(c_1) + v(1-L) + \beta u\big(F_2(K)\big) + \lambda\big[F_1(L) - c_1 - K\big]$$

- First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial c_1} = 0: \quad u'(c_1) - \lambda = 0$$
$$\frac{\partial \mathcal{L}}{\partial L} = 0: \quad -v'(1-L) + \lambda F_1'(L) = 0$$
$$\frac{\partial \mathcal{L}}{\partial K} = 0: \quad \beta u'\big(F_2(K)\big)F_2'(K) - \lambda = 0$$

Two optimality conditions

- Substitute $\lambda = u'(c_1)$ into the other two FOCs:

(1) optimal labor supply

$$v'(l) = F'_1(L)\,u'(c_1)$$

(2) Euler equation

$$u'(c_1) = \beta F'_2(K)\,u'(c_2)$$

- Both say: marginal cost = marginal benefit.

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- Both say: marginal cost = marginal benefit.
- *Prices*: firms pay factors their marginal products, and capital is fully used up in period 2 (as if $\delta = 1$):

$$w = F_1'(L) = \frac{v'(l)}{u'(c_1)} \qquad 1 + r = F_2'(K) = \frac{u'(c_1)}{\beta u'(c_2)}$$

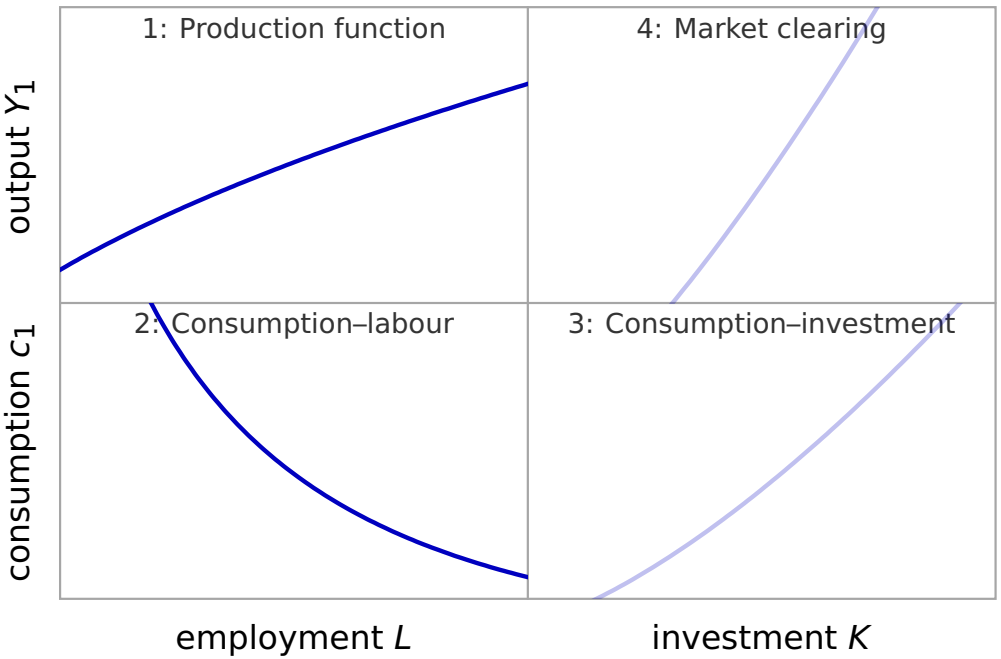
The model in four equations

- Use the constraints to substitute out $c_2 = F_2(K)$ and $l = 1 - L$:

(1)	$Y_1 = F_1(L)$	*production function*
(2)	$v'(1-L) = F_1'(L)u'(c_1)$	*optimal labour supply*
(3)	$u'(c_1) = \beta F_2'(K)\,u'(F_2(K))$	*Euler equation*
(4)	$Y_1 = c_1 + K$	*market clearing*

- 4 equations in 4 unknowns: Y_1, c_1, K, L.
- We study the solution in two ways:
  - **graphically**, for general functions u, v, F_1, F_2;
  - **algebraically**, for specific functional forms.

The four curves (1/2)

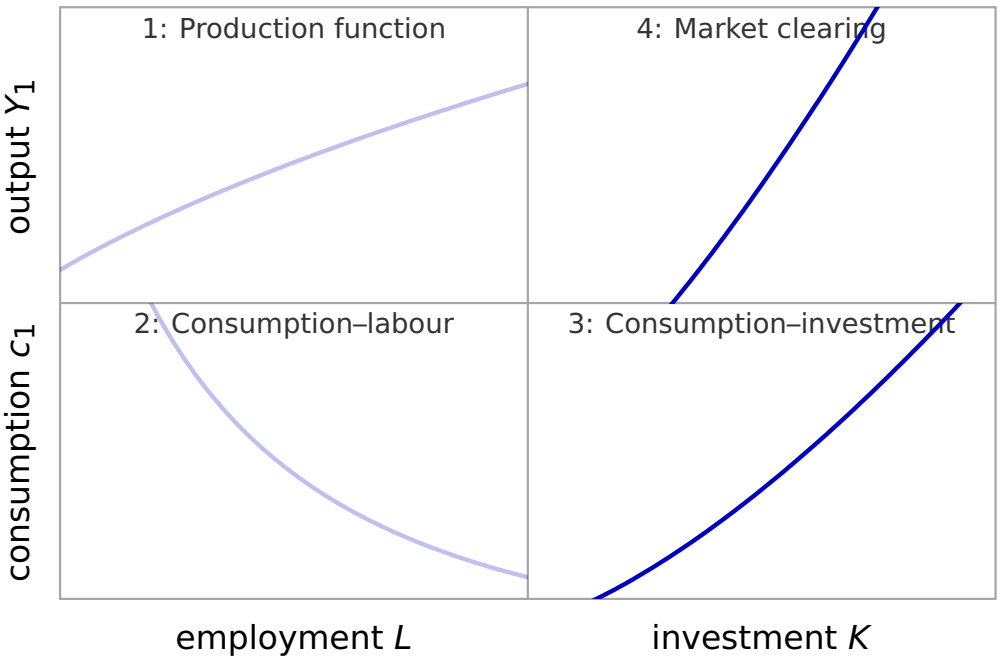


Panel 1 – production function: upward sloping. More work, more output.

Panel 2 – labour supply: downward sloping. At lower consumption level, $u'(c)$ is higher so household has an incentive to increase work.

Wealth effect: *other things equal*, households work more when they feel poorer.

The four curves (2/2)



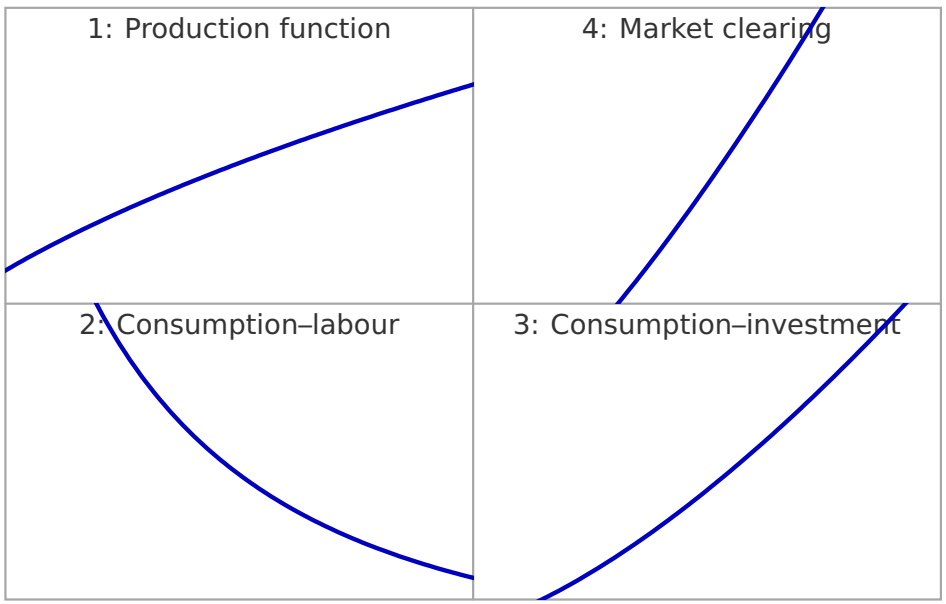
Panel 3 – Euler equation: upward sloping. At higher consumption level, $u'(c_1)$ is lower so household is willing to invest more.

Consumption smoothing: *other things equal*, households save more when they feel richer today.

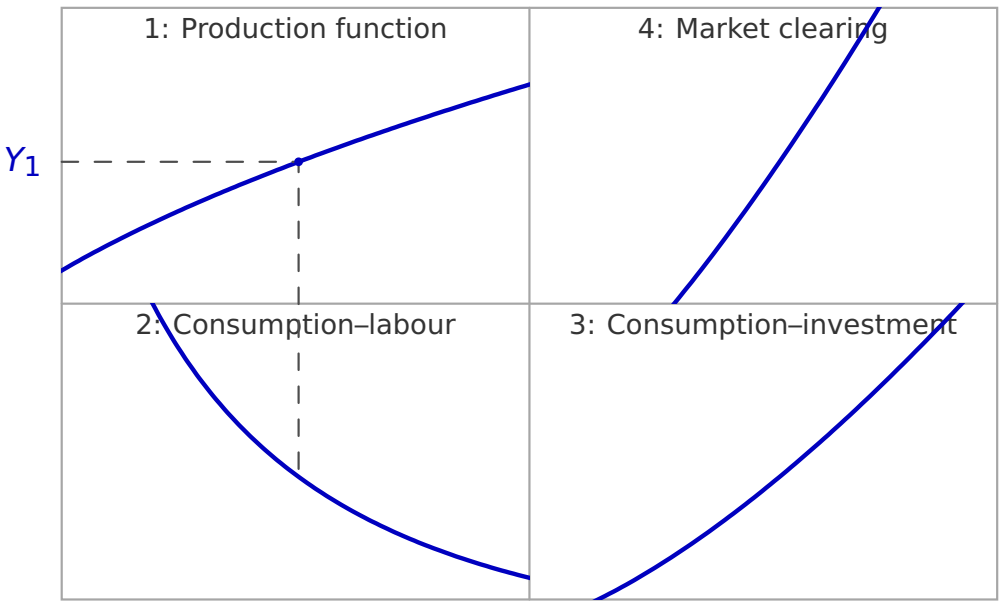
Panel 4 – market clearing
$Y_1 = c_1 + K$: upward sloping. Higher investment requires higher output.

Solution = the set of values for Y_1, c_1, K, L that satisfies all four equations (lie on all four curves).

Solving the model graphically

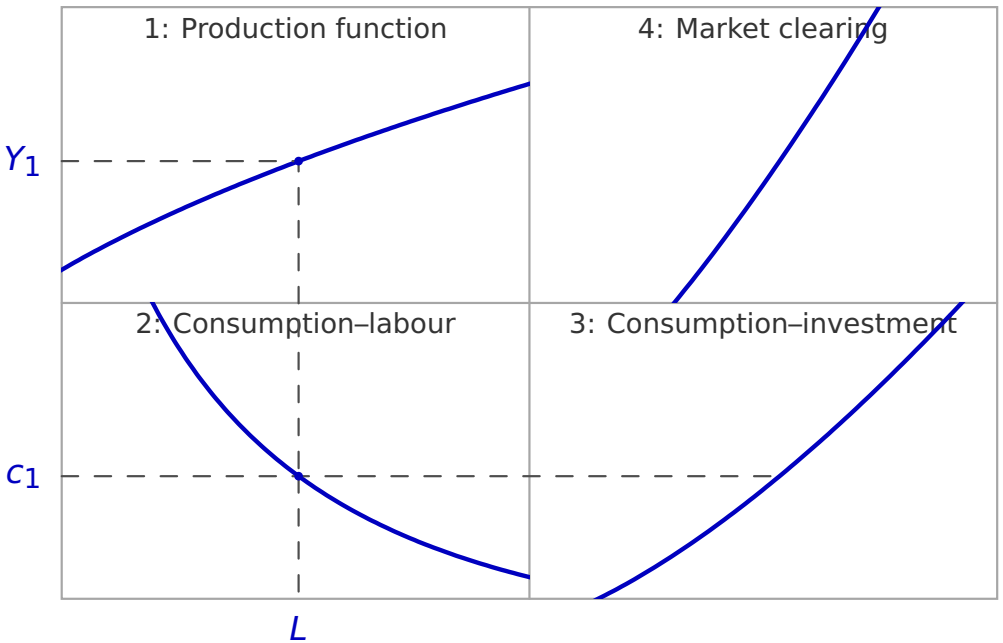


Solving the model graphically



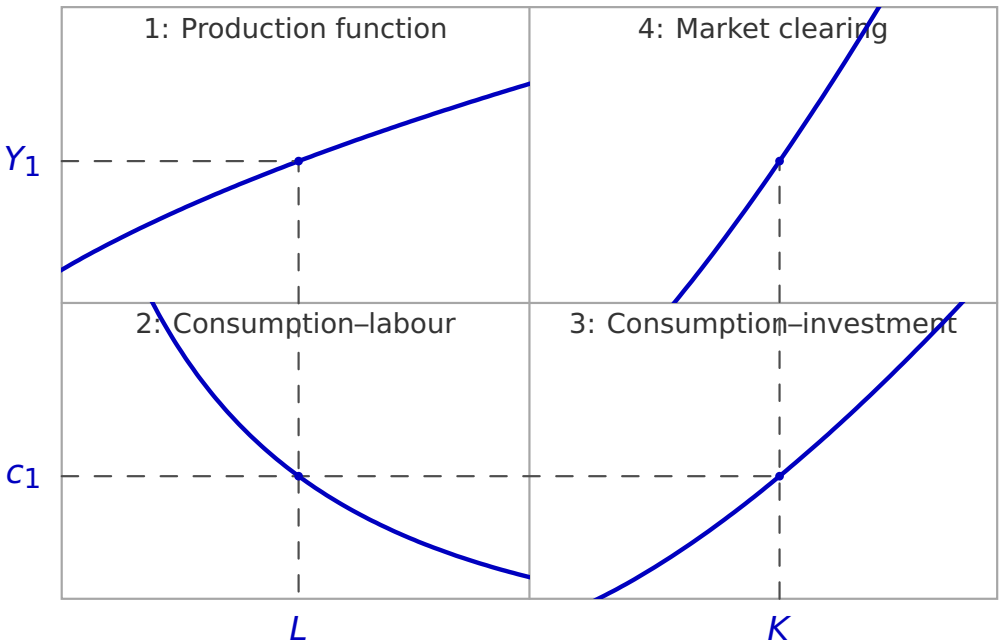
1. *Guess* Y_1. The production function (panel 1) tells us the L needed to produce it.

Solving the model graphically



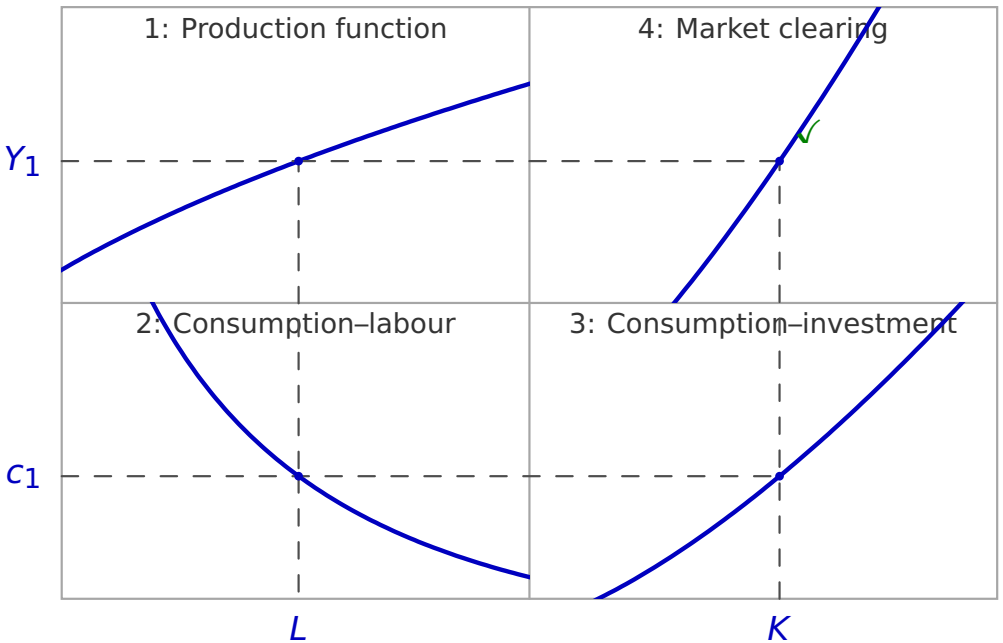
1. *Guess* Y_1. The production function (panel 1) tells us the L needed to produce it.
2. Given L, the labor supply curve (panel 2) gives c_1.

Solving the model graphically



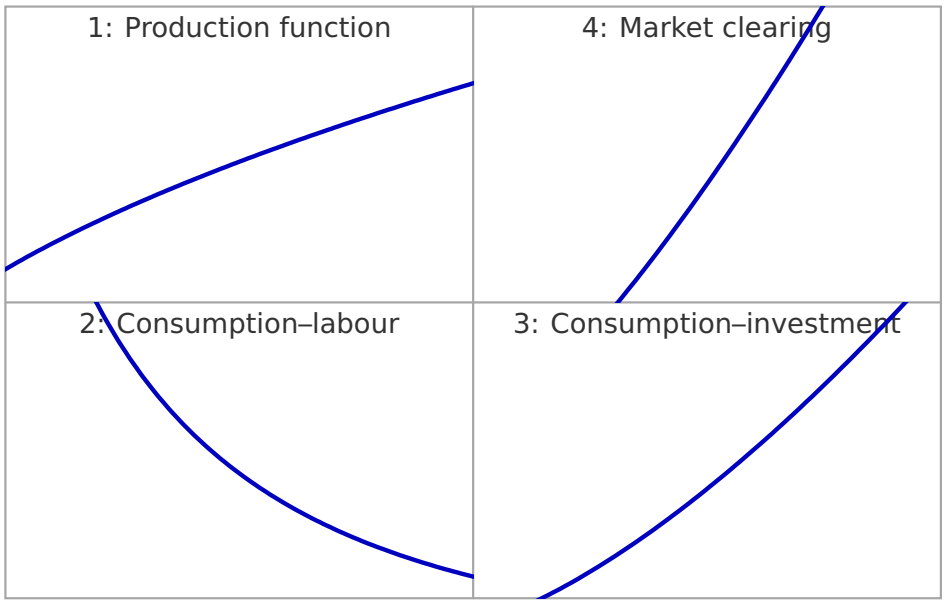
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2. Given L, the labor supply curve (panel 2) gives c_1.
3. Given c_1, the Euler equation (panel 3) gives K.

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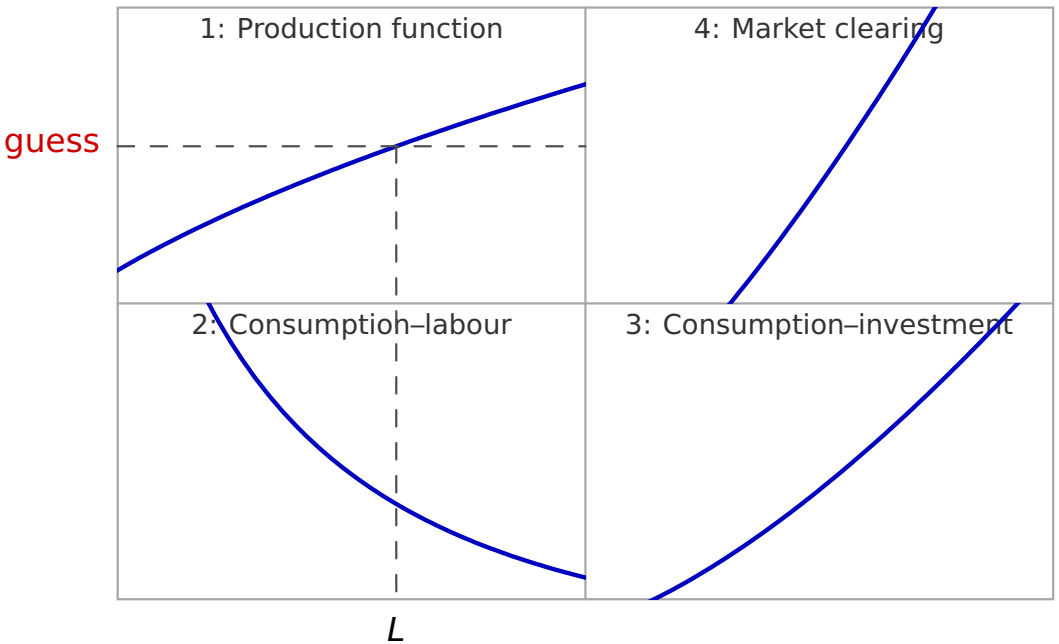


1. *Guess* Y_1. The production function (panel 1) tells us the L needed to produce it.
2. Given L, the labor supply curve (panel 2) gives c_1.
3. Given c_1, the Euler equation (panel 3) gives K.
4. Check in panel 4: does $c_1 + K$ equal the Y_1 we guessed? Yes ⇒ all four equations hold: **equilibrium**.

What if the guess is wrong?

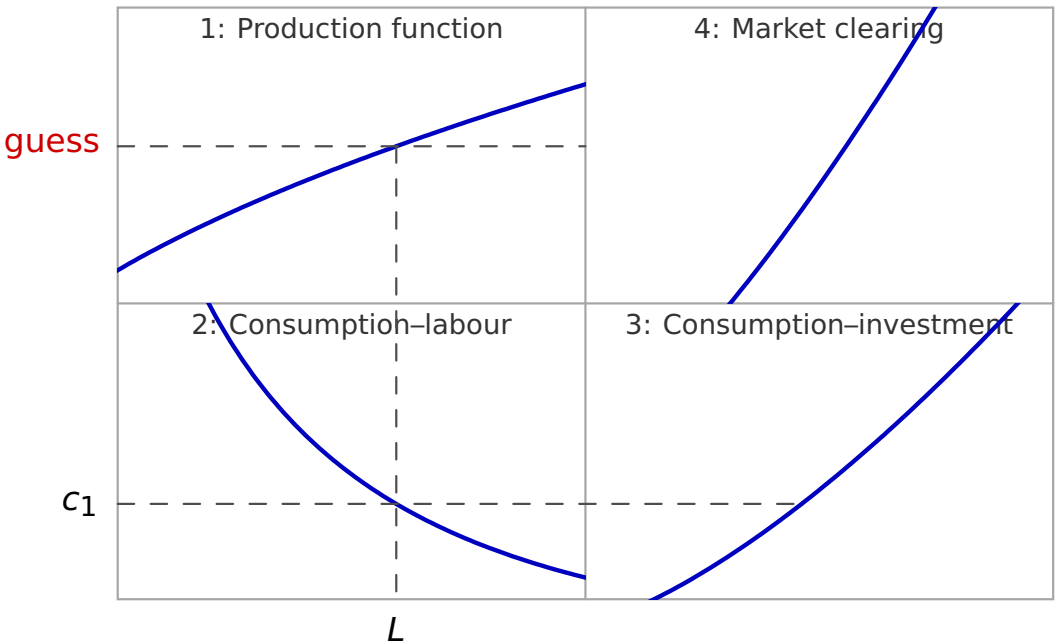


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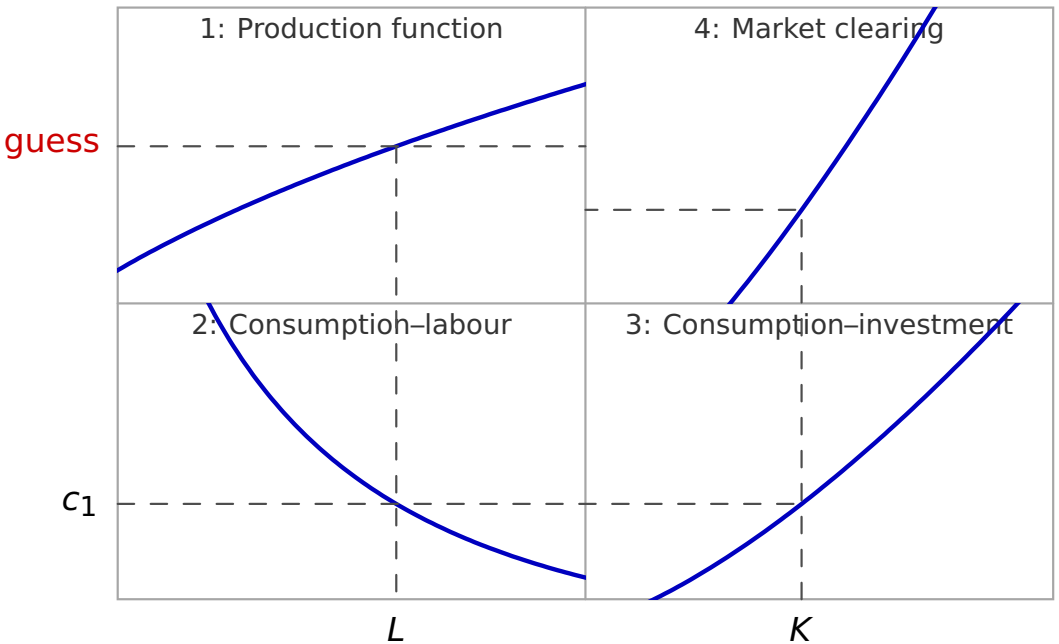
- Guess a Y_1 that is too high ⇒ a lot of labour is needed (panel 1);

What if the guess is wrong?



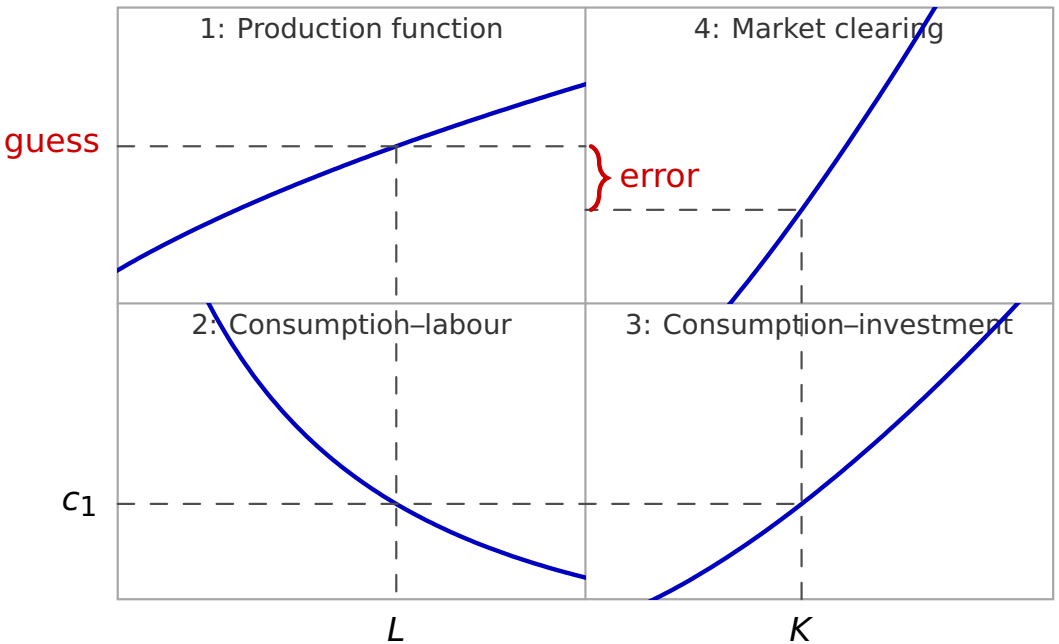
- Guess a Y_1 that is too high ⇒ a lot of labour is needed (panel 1);
- high L ⇒ low c_1: households work a lot only if they feel poor (panel 2);

What if the guess is wrong?



- Guess a Y_1 that is too high ⇒ a lot of labour is needed (panel 1);
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- low c_1 ⇒ low K (panel 3);

What if the guess is wrong?



- Guess a Y_1 that is too high ⇒ a lot of labour is needed (panel 1);
- high L ⇒ low c_1: households work a lot only if they feel poor (panel 2);
- low c_1 ⇒ low K (panel 3);
- low c_1 and low K ⇒ $c_1 + K$ is *below* the guess: inconsistent! Lower the guess and try again.

(The order does not matter: we could also go 1-4-3-2.)

Solving the model algebraically: functional forms

- To get an explicit solution we need specific functions:

$$u(c) = \ln c, \qquad v(l) = b \ln l, \qquad F_1(L) = L^{1-\alpha}, \qquad F_2(K) = K^{\alpha}$$

- Marginal utilities and products: $u'(c) = \frac{1}{c}$, $v'(l) = \frac{b}{l}$, $F_1'(L) = (1-\alpha)L^{-\alpha}$, $F_2'(K) = \alpha K^{\alpha-1}$
- The four equations become:

$$\begin{aligned}
&(1) \quad Y_1 = L^{1-\alpha} && \textit{production function} \\
&(2) \quad \frac{b\,c_1}{1-L} = (1-\alpha)L^{-\alpha} && \textit{optimal labor supply} \\
&(3) \quad \frac{1}{c_1} = \beta\,\alpha K^{\alpha-1}\,\frac{1}{K^{\alpha}} = \frac{\alpha\beta}{K} && \textit{Euler equation} \\
&(4) \quad Y_1 = c_1 + K && \textit{market clearing}
\end{aligned}$$

The full solution

- Two steps (details in the appendix):
  - Euler equation with log utility $\Rightarrow K = \alpha\beta c_1$: households invest a constant fraction $s = (\alpha\beta)/(1+\alpha\beta)$ of output;
  - put $c_1 = (1-s)Y_1$ into (2) and solve for L.

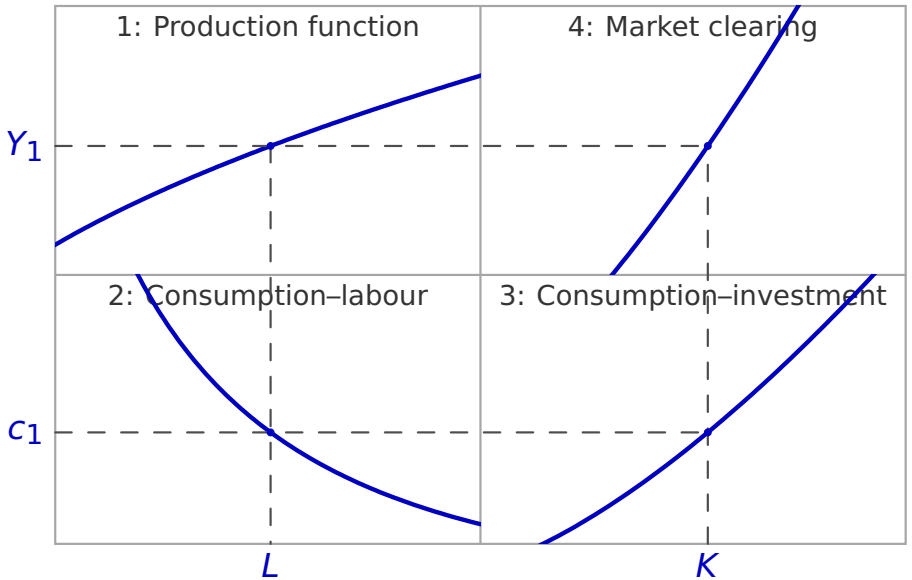
$$s = \frac{\alpha\beta}{1+\alpha\beta} \qquad L^* = \frac{1-\alpha}{(1-\alpha)+b(1-s)}$$

$$Y_1^* = (L^*)^{1-\alpha} \qquad c_1^* = (1-s)Y_1^* \qquad K^* = sY_1^* \qquad c_2^* = (K^*)^{\alpha}$$

$$w^* = F_1'(L^*) = (1-\alpha)\frac{Y_1^*}{L^*} \qquad 1+r^* = F_2'(K^*) = \alpha(K^*)^{\alpha-1}$$

- Every endogenous variable depends on the parameters α, β, b only: more patient households ($\beta \uparrow$) save more; households who value leisure more ($b \uparrow$) work less and produce lower output.

A temporary productivity shock



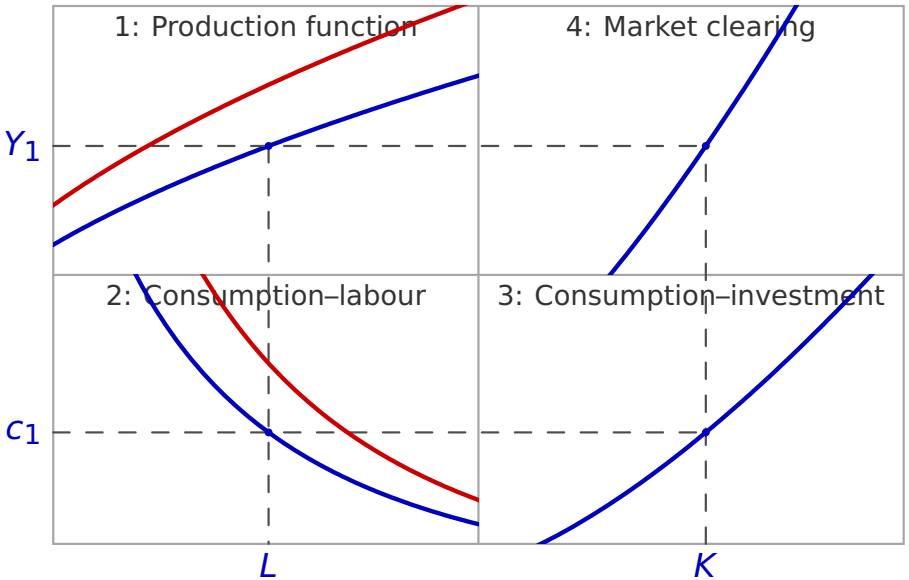
- Shift in period-1 productivity, with period 2 unchanged.

$$Y_1 = AF_1(L), \quad A > 1$$

- A enters two equations:

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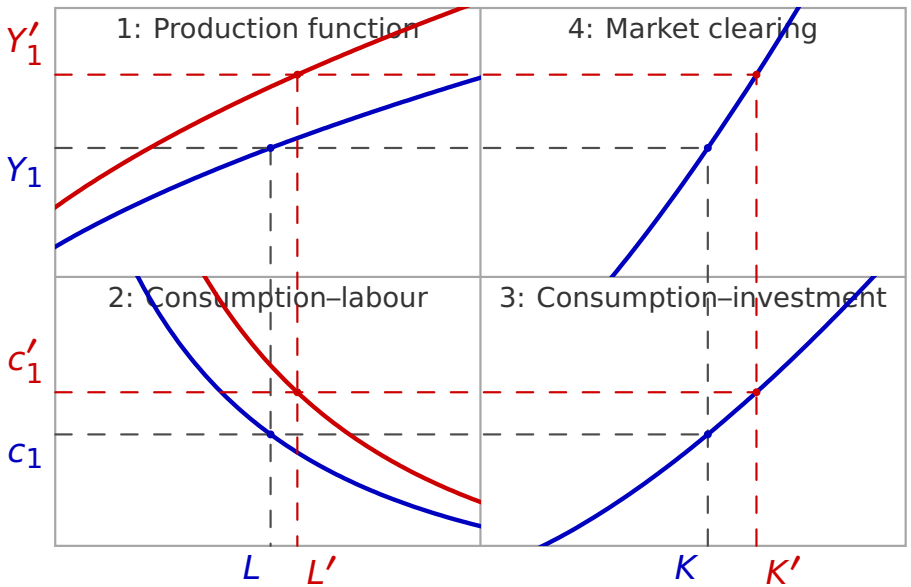
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- Panel 1 shifts up: more output for any L. Panel 2 shifts right: work is more rewarding.

A temporary productivity shock



With log utility and $F_1(L) = L^{1-\alpha}$:

$$Y_1^* = A\,(L^*)^{1-\alpha}$$
$$c_1^* = (1-s)\,A\,(L^*)^{1-\alpha}$$
$$K^* = sA\,(L^*)^{1-\alpha}$$
$$L^* = \frac{1-\alpha}{(1-\alpha)+b\,(1-s)} \quad \text{(no } A\text{)}$$

- Shift in period-1 productivity, with period 2 unchanged.

$$Y_1 = AF_1(L), \quad A > 1$$

- A enters two equations:

$$Y_1 = AF_1(L), \quad \frac{v'(1-L)}{u'(c_1)} = AF_1'(L)$$

- Panel 1 shifts up: more output for any L. Panel 2 shifts right: work is more rewarding.
- New equilibrium: Y_1, c_1, K all rise.
- L: the shift (work more) vs the movement along the curve as c_1 rises (work less).

  Substitution vs wealth effect.

  In the diagram L rises; with log utility (left) they cancel.

Temporary productivity shocks and behavior

After a temporary productivity improvement, the household wants to:

1. **consume more**: it feels wealthier (*wealth effect*);
2. **invest more**: to spread the extra consumption over both periods (*smoothing*);
3. **perhaps work more**: time spent working is unusually productive now (*substitution effect*);
4. **but perhaps enjoy more leisure**: it feels wealthier (*wealth effect*).

- 1 and 2 are unambiguous; 3 and 4 go in opposite directions; which one wins depends on preferences (σ: see appendix).

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- 1 and 2 are unambiguous; 3 and 4 go in opposite directions; which one wins depends on preferences (σ: see appendix).
- If substitution effect wins, Y, L, c_1 and investment all move together: looks like the business cycles we observe
- Productivity shocks as the source of business cycles:
  - *booms*: households choose to work more when productivity (and the wage) is temporarily high;
  - *recessions*: households *choose* to work less when it is temporarily low.

The plan

1. A simple two-period RBC model (Kurlat);
2. **The full, infinite-horizon RBC model (Romer, ch. 5);**
3. Assessment: calibration, policy implications, extensions and criticisms.

Assumptions about production

- Time is discrete and runs from 0 to infinity
- representative firm; Cobb-Douglas production function:

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}, \quad 0 < \alpha < 1$$

- Evolution of K:

$$K_{t+1} = K_t + I_t - \delta K_t$$

- Factor remunerations:

$$w_t = MPL_t \qquad r_t = MPK_t - \delta$$

Behavioral assumptions

- Fixed large number of identical infinitely-lived households;
- Representative household maximizes *expected value* of

$$U = \sum_{t=0}^{\infty} e^{-\rho t} u(c_t, 1-\ell_t) \frac{N_t}{H}$$

- Instantaneous utility:

$$u_t = \ln c_t + b \ln(1-\ell_t), \quad b > 0$$

  - here ℓ_t is *labour* and $1-\ell_t$ is leisure (in Kurlat, l was leisure);
  - $e^{-\rho}$ plays the role of Kurlat's β.

- H fixed but (exogenous) population growth:

$$\ln N_t = \bar{N} + nt, \quad n < \rho$$

Technological change and government

- *Technological progress:*

$$\ln A_t = \bar{A} + gt + \tilde{A}_t$$

$$\tilde{A}_t = \rho_A \tilde{A}_{t-1} + \epsilon_{A,t}, \qquad -1 < \rho_A < 1$$

  - $\epsilon_{A,t}$ = uncorrelated zero-mean disturbance (*white-noise*)
  - ρ_A makes the shocks persistent (by assumption).

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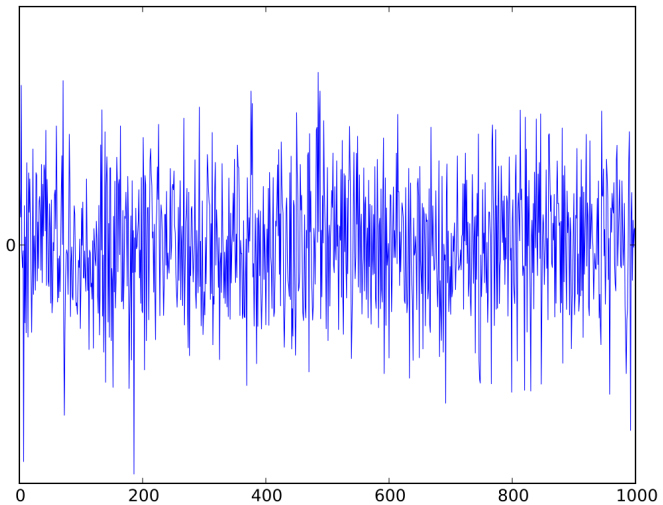
- *Government purchases:*

$$\ln G_t = \bar{G} + (n+g)t + \tilde{G}_t$$

$$\tilde{G}_t = \rho_G \tilde{G}_{t-1} + \epsilon_{G,t}, \quad -1 < \rho_G < 1$$

- (same logic)

Example of white-noise ϵ_t term:



- Value in each period is a random draw with mean zero, independent of previous periods (uncorrelated);
- Persistence is added through the ρ terms;

Euler Equation

- Formal derivation more cumbersome with uncertainty and infinite horizon, but fundamentally the same as in the two-periods model.
- 'intuitive derivation': Optimization requires that a marginal reallocation of C from t to $t+1$ does not affect expected utility
- Δc decrease in consumption per head in t
  $\Rightarrow e^{-n}(1+r_{t+1})\Delta c$ increase in consumption per head in $t+1$.
- Equating utility cost and (expected) utility benefit:

$$\left(e^{-\rho t}\frac{N_t}{H}\frac{1}{c_t}\right)\Delta c = E_t\left[\left(e^{-\rho(t+1)}\frac{N_{t+1}}{H}\frac{1}{c_{t+1}}\right)\frac{1+r_{t+1}}{e^n}\Delta c\right]$$

- Rewrite as

$$\frac{1}{c_t} = e^{-\rho}E_t\left[\frac{1+r_{t+1}}{c_{t+1}}\right]$$

Optimal labor supply

- 'Intuitive derivation' again
- Optimization requires that a marginal increase in ℓ_t, using resulting income to raise c_t, does not affect expected utility
  - $$\left(e^{-\rho t}\frac{N_t}{H}\frac{b}{1-\ell_t}\right)\Delta\ell = \left(e^{-\rho t}\frac{N_t}{H}\frac{1}{c_t}\right)w_t\Delta\ell$$
  - *marginal disutility of* $\ell_t = w_t \times$ *marginal utility of* c_t
- Rewrite as
$$\frac{c_t}{1-\ell_t} = \frac{w_t}{b}$$
  - consumption/leisure ratio is a positive function of the wage.
  - as in 2-periods model, other things equal, there is a negative relation between consumption and labour supply.

Solutions for the infinite-horizon model

- The same two optimality conditions, now holding *in every period*.
- Given random shocks, household can't choose deterministic future paths for C and L at time 0, but needs to 'recalculate' in each period based on shocks realization
  - (like Google Maps when there are changes in traffic).
- The infinite-horizon RBC model cannot in general be solved analytically (mix of linear and log-linear equations) – numerical solutions are used.
- We will focus on the intuition without going into the technicalities.

Solving the infinite-horizon RBC model (intuition)

- Take a log-linear approximation, and apply the method of undetermined coefficients. Four steps:

1. Approximate the model around its balanced growth path (Taylor's theorem): work with % deviations from trend, $\tilde{X}_t = \ln X_t - \ln X_t^*$.

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1. Approximate the model around its balanced growth path (Taylor's theorem): work with % deviations from trend, $\tilde{X}_t = \ln X_t - \ln X_t^*$.
2. Guess that in the log-linearised solution, each endogenous variable will be a log-linear function of the state of the economy (K, A, G):

$$\tilde{C}_t \approx a_{CK}\tilde{K}_t + a_{CA}\tilde{A}_t + a_{CG}\tilde{G}_t \qquad \tilde{L}_t \approx a_{LK}\tilde{K}_t + a_{LA}\tilde{A}_t + a_{LG}\tilde{G}_t$$

Solving the infinite-horizon RBC model (intuition)

- Take a log-linear approximation, and apply the method of undetermined coefficients. Four steps:

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3. Use the two (log-linearised) optimality conditions to pin down the a's as a function of the model parameters
4. Assume a shock and simulate the reaction of the economy to it (*impulse response function*).

- For more details: appendix here; Romer Ch 5; Campbell, 1994.

What the solution looks like

Assumptions about parameter values:

- a period is a quarter;
- $\alpha = \frac{1}{3}$, $g = 0.5\%$, $n = 0.25\%$, $\delta = 2.5\%$;
- $\rho_A = \rho_G = 0.95$; $\bar{G}$: $G/Y = 0.2$;
- ρ: $r^\star = 1.5\%$; b: $\ell^\star = \frac{1}{3}$.

$\rightarrow$

Resulting coefficients:

- $a_{LA} \approx 0.35$, $a_{LK} \approx -0.31$
- $a_{CA} \approx 0.38$, $a_{CK} \approx 0.59$
- $b_{KA} \approx 0.08$, $b_{KK} \approx 0.95$

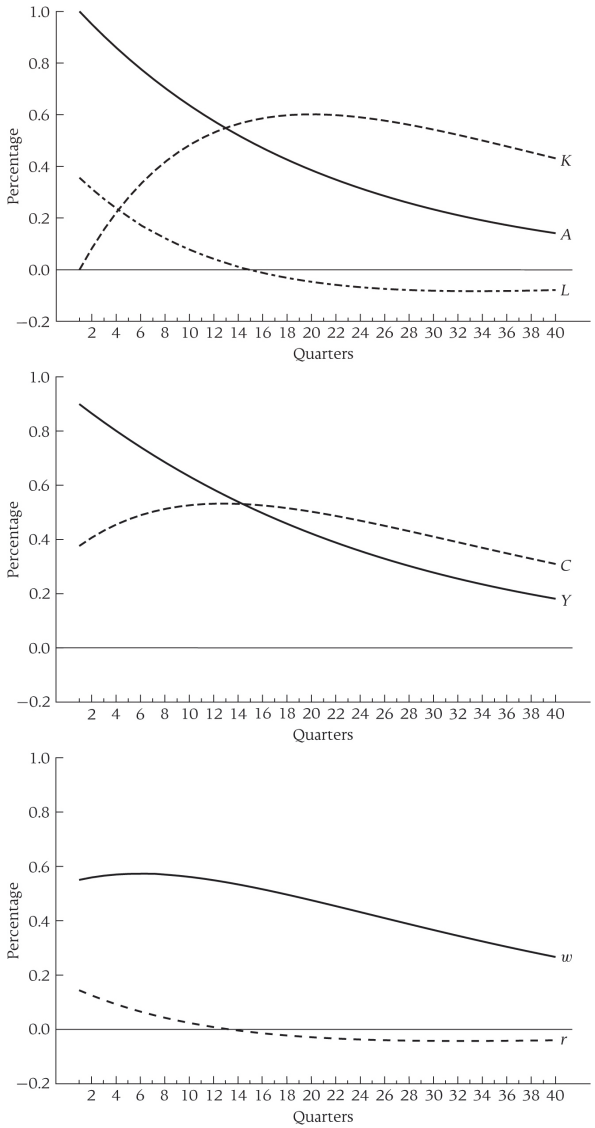
(where $\tilde{K}_{t+1} \approx b_{KK}\tilde{K}_t + b_{KA}\tilde{A}_t + b_{KG}\tilde{G}_t$)

Reading the numbers – a 1% technology shock:

- raises employment by 0.35% and consumption by 0.38% on impact;
- raises output by about 0.9% on impact (from production function, with 1% increase in A + 0.35% increase in L, with K unchanged);
- raises next quarter's capital by 0.08%.

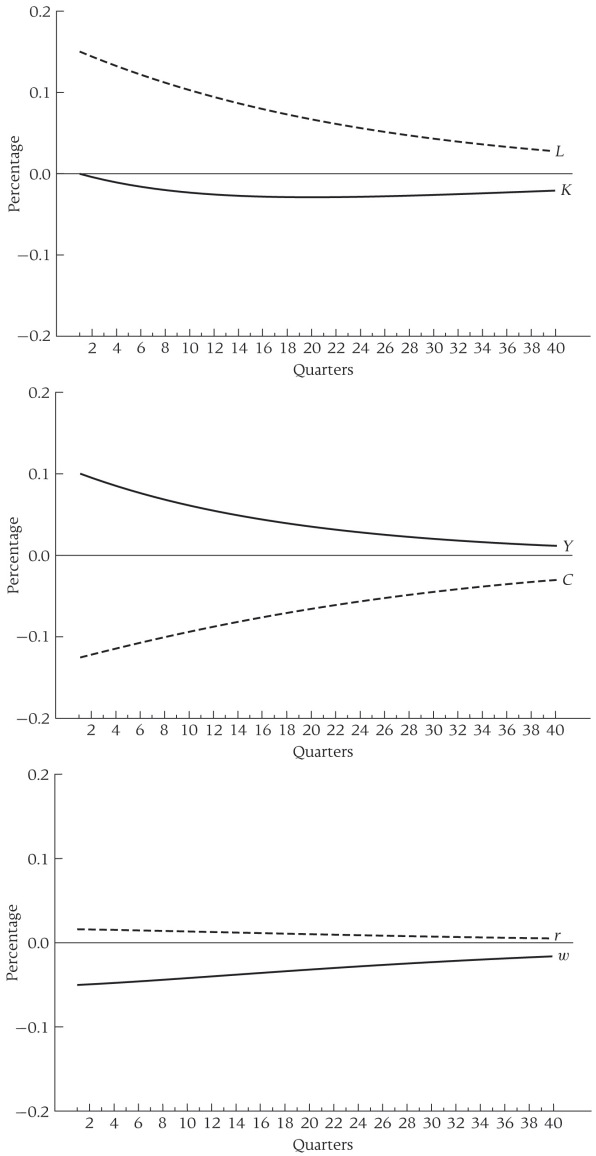
$a_{LK} < 0$: more capital makes households richer, so they work less.

Effect of a positive technology shock



- *A* shock is temporary but persistent (by assumption).
- *Wealth effect*: households are richer ⇒ $C \uparrow$, $L \downarrow$
- *Substitution effects*:
  - high *MPL* ⇒ $L \uparrow$
  - high *MPK* ⇒ $L \uparrow$ $C \downarrow$
- In this case, substitution dominates: *L* and saving rate *s* rise on impact
- *C* still rises but much less than *Y* ⇒ investment more volatile than consumption (as in the data).
- *w* and *r* rise, but not by much
  - increase in *L* and *K* partly offset the positive effect on MPL and MPK.

Effect of a gov't purchases shock



- G rises temporarily (but persistently), financed by lump-sum taxes.
- The government absorbs output, households are poorer & less output available for C and I.
- *Negative wealth effect:* $C \downarrow$ and $L \uparrow$.
- *Consumption smoothing:* The rise in G is temporary $\Rightarrow$ households also reduce investment ($I \downarrow$) to smooth out the fall in C.
- C falls by about one-third of the rise in G.
- More L with roughly the same $K \Rightarrow w$ falls slightly, r rises.
- Output rises slightly (by about half the rise in G) because of higher labor supply.
- Is this a good way to model government spending? It grabs output without doing anything with it?

The plan

1. A simple two-period RBC model (Kurlat);
2. The full, infinite-horizon RBC model (Romer, ch. 5);
3. **Assessment: calibration, policy implications, extensions and criticisms.**

Calibration of RBC models

The procedure (Kydland and Prescott, 1982; Prescott, 1986):

1. build the full (infinite-horizon) model;
2. set parameter values using *micro* evidence (e.g. household data on labour supply);
3. measure productivity shocks as Solow residuals (growth accounting, quarter by quarter);
4. simulate the model hit by shocks of that size;
5. compare volatilities and correlations in the model with those in the data.

TABLE 5.4 A calibrated real-business-cycle model versus actual data

	U.S. data	Baseline real-business-cycle model
σ_Y	1.92	1.30
σ_C/σ_Y	0.45	0.31
σ_I/σ_Y	2.78	3.15
σ_L/σ_Y	0.96	0.49
Corr(L,Y/L)	−0.14	0.93

Source: Hansen and Wright (1992).

Successes: output volatility about 2/3 of the data; C less and I more volatile than Y.

Failures: hours too smooth (σ_L/σ_Y 0.49 vs 0.96); hours and productivity far too correlated (0.93 vs −0.14).

Main extensions to the baseline RBC model

Indivisible labor:

- hours worked are a discrete variable (either work full-time or zero);
- raises σ_Y and σ_L/σ_Y

Multiple sectors

- effect of sector-specific technological shocks (hypothesis that sector-specific shocks can significantly explain aggregate fluctuations).

Distortionary taxation

- Distortionary taxes ($T_t = \tau_t Y_t$) to finance government purchases;
- they can make people work less (intertemporal substitution), making fiscal expansions *contractionary*;

Policy implications

- If the RBC model is right, the First Welfare Theorem holds: no planner – however powerful – could do better than the market.
- Fluctuations are the *efficient* response to changes in productivity:
  - when productivity is high it is a good time to produce, so people *should* work more;
  - when it is low, people *should* enjoy more leisure.
- No stabilisation policy: Monetary policy cannot affect real variables (no money); fiscal policy can (slightly) move output, but stabilising would *reduce* welfare.
- There is no money in the model: monetary policy is irrelevant at best.

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- No stabilisation policy: Monetary policy cannot affect real variables (no money); fiscal policy can (slightly) move output, but stabilising would *reduce* welfare.
- There is no money in the model: monetary policy is irrelevant at best.
- Observing fluctuations does not by itself prove that something is wrong. Any case for stabilisation must explain *why* fluctuations are inefficient.
- … but is the RBC model right?

Why (almost) no one believes RBC theory (1/3): the shocks

- The model needs technology shocks of about 0.7% *per quarter*. Recessions as technological regress? Why don't we read about them in the newspaper?
  - "*[RBC models] attribute fluctuations in aggregate variables to imaginary causal forces that are not influenced by the action that any person takes.*"(P. Romer, 2016)
  - Consumption is micro-founded but productivity is not!
- The Solow residual (or other productivity measures) are not clean measures of technology: short-run fluctuations probably reflect demand-driven changes in utilization rates.
  - they move with the Great Depression, the WWII boom, military spending, oil prices...
- On a slow night a restaurant has the same staff (L) and kitchen (K) but serves fewer meals.
  - measured productivity falls – but because customers did not show up (demand), not because the technology got worse;
  - varying utilisation of capital and labour makes measured productivity fluctuate with demand.

Why (almost) no one believes RBC theory (2/3): the mechanism

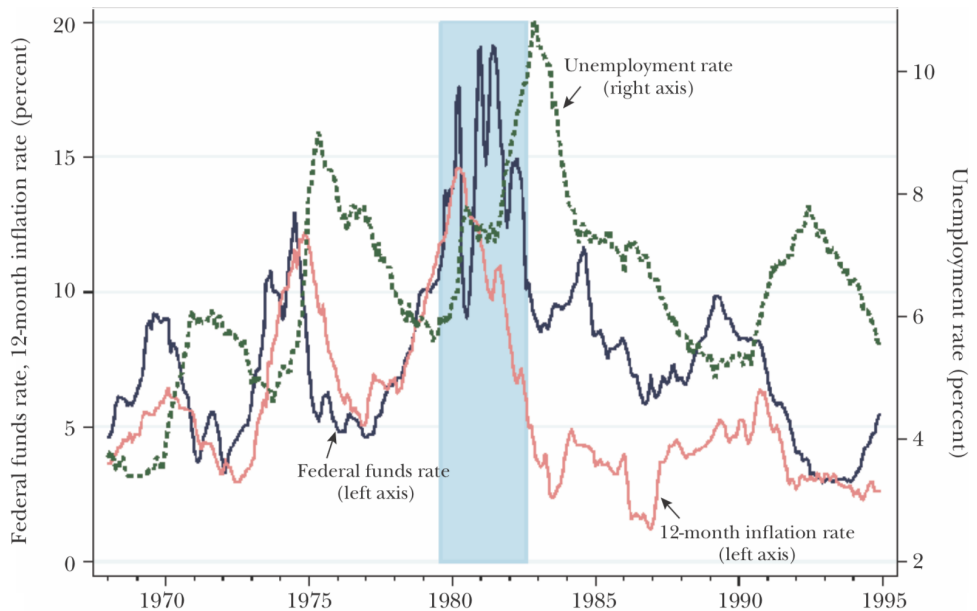
- Large employment fluctuations require households to be very willing to shift work across time; micro studies find this elasticity is low.
- No involuntary unemployment: employment falls only because people *choose* to work less.
  - The Great Depression as a Great Vacation?
  - In reality, recessions are associated with significant unemployment
- When economists isolate genuine technology shocks, hours worked *fall* after a positive shock (e.g. Galí; Basu, Fernald and Kimball) – the opposite of the RBC prediction, but what sticky-price models predict.
- No internal propagation mechanisms: fluctuations are persistent only because the shocks themselves are assumed to be (Cogley and Nason, 1995).

Why (almost) no one believes RBC theory (3/3): money matters

- Deviations from perfect and complete markets are so pervasive that it seems incredible that they don't have substantial macro effects.
- A truckload of evidence that monetary policy affects real variables. E.g. the 'Volcker recession' (1979-1982):

Figure 2

Federal Funds Rate, Inflation, and Unemployment from 1965 to 1995



Note: The figure plots the federal funds rate (dark solid line, left axis), the 12-month inflation rate (light solid line, left axis), and the unemployment rate (dashed line, right axis). The Volcker disinflation period is the shaded bar (August 1979 to August 1982).

Takeaways

- RBC theory: fluctuations as the *optimal* response of competitive markets to real shocks.
- *Two-period model* isolates the core mechanisms in the simplest setting
  - wealth vs substitution effects on labour supply;
  - tech shocks shift the consumption–labour tradeoff and can make Y, L, C and I move together.
- *Infinite-horizon model*: the same two optimality conditions every period, plus
  - capital accumulation and persistent shocks → dynamics;
  - quantitative predictions → calibration.
- *Assessment*: matches some co-movements, but implausible shocks, unrealistic labour market, wild implications (zero role for money and monetary policy; zero involuntary unemployment)
- Lasting legacy: the *method*. New-Keynesian models = RBC structure + nominal rigidities and imperfect competition.

Appendix

Technical details of the two-period and infinite-horizon models

(optional: for those who want to see the derivations)

Step 1: use the Euler equation to find the saving rate

- Equations (1)–(4) as on the 'functional forms' slide. From (3): $K = \alpha\beta\, c_1$

- Substitute into (4): $Y_1 = c_1 + \alpha\beta c_1 = (1+\alpha\beta)\, c_1$, so

$$c_1 = \frac{1}{1+\alpha\beta}\, Y_1, \qquad K = \frac{\alpha\beta}{1+\alpha\beta}\, Y_1$$

- The share of output that is invested (the saving rate) is:

$$s \equiv \frac{K}{Y_1} = \frac{\alpha\beta}{1+\alpha\beta}$$

- s depends only on parameters – not on L or on how much is produced:
  - more patient households ($\beta \uparrow$) save more;
  - if capital matters more in production ($\alpha \uparrow$) households save more;
  - with log utility, households always save the same fraction of their income.

Step 2: use the labour condition to find employment

- Use $c_1 = (1-s)Y_1 = (1-s)L^{1-\alpha}$ in (2):

$$\frac{b(1-s)L^{1-\alpha}}{1-L} = (1-\alpha)L^{-\alpha}$$

- Multiplying both sides by L^{α}:

$$b(1-s)\frac{L}{1-L} = 1-\alpha$$

- Solving for L:

$$L^* = \frac{1-\alpha}{(1-\alpha)+b(1-s)}$$

- Employment also depends only on parameters:
  - households who value leisure more ($b \uparrow$) work less.

A productivity shock: the algebra

- Redo the solution with $Y_1 = AL^{1-\alpha}$, $A > 1$:
  - s is unchanged: A does not enter (3) or (4);
  - in (2), both c_1 and the *MPL* rise in proportion to A, so A cancels: L^* **unchanged!**
  - Y_1^*, c_1^*, K^* rise in proportion to A; c_2^*, w^* rise too; $1 + r^*$ falls.
- Why? With log utility, the wealth effect (4) and the substitution effect (3) on labour supply *exactly cancel*.

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  - Y_1^*, c_1^*, K^* rise in proportion to A; c_2^*, w^* rise too; $1 + r^*$ falls.
- Why? With log utility, the wealth effect (4) and the substitution effect (3) on labour supply *exactly cancel*.
- More generally, with $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ (log utility is the case $\sigma = 1$; $1/\sigma$ = willingness to shift consumption over time):

	stronger effect	employment after $A \uparrow$
$\sigma < 1$	substitution	$L \uparrow$ (the case in the lecture diagram)
$\sigma = 1$	neither: they cancel	L unchanged
$\sigma > 1$	wealth	$L \downarrow$

- Lesson: whether employment rises in a boom is a horse race between substitution and wealth effects – a recurring theme in the infinite-horizon model.

(1) Euler Equation

- 'Intuitive derivation'
- Optimization requires that a marginal reallocation of C from t to $t+1$ does not affect expected utility
- Δc decrease in consumption per head in t
  $\Rightarrow e^{-n}(1+r_{t+1})\Delta c$ increase in consumption per head in $t+1$.
- Equating utility cost and (expected) utility benefit:

$$\left(e^{-\rho t}\frac{N_t}{H}\frac{1}{c_t}\right)\Delta c = E_t\left[\left(e^{-\rho(t+1)}\frac{N_{t+1}}{H}\frac{1}{c_{t+1}}\right)\frac{1+r_{t+1}}{e^n}\Delta c\right]$$

- Rewrite as

$$\frac{1}{c_t} = e^{-\rho}E_t\left[\frac{1+r_{t+1}}{c_{t+1}}\right]$$

(2) Optimal labor supply

- 'Intuitive derivation' again
- Optimization requires that a marginal increase in ℓ_t, using resulting income to raise c_t, does not affect expected utility
  - $$\left(e^{-\rho t}\frac{N_t}{H}\frac{b}{1-\ell_t}\right)\Delta\ell = \left(e^{-\rho t}\frac{N_t}{H}\frac{1}{c_t}\right)w_t\Delta\ell$$
  - *marginal disutility of* $\ell_t = w_t \times$ *marginal utility of* c_t
- Rewrite as
$$\frac{c_t}{1-\ell_t} = \frac{w_t}{b}$$
  - consumption/leisure ratio is a positive function of the wage.

Solving the special case ($\delta = 1$, no G)

- With only circulating capital ($\delta = 1$) and no government:

$$K_{t+1} = I_t = Y_t - C_t \qquad r_t = \alpha \left(\frac{A_t L_t}{K_t} \right)^{1-\alpha} - 1$$

- two state variables: K and A; two endogenous variables: C and L;
- conjecture: saving rate s and labour supply ℓ constant in equilibrium;
- Strategy:
  1. use equilibrium conditions for household behavior to find $\hat{s}$ and $\hat{\ell}$;
  2. Study equilibrium behavior of aggregates;

Step 1: use Euler Equation to find $\hat{s}$.

- Rewrite EE in terms of s using $c_t = (1-s_t)\frac{Y_t}{N_t}$, and take logs:

$$-\ln\left[(1-s_t)\frac{Y_t}{N_t}\right] = -\rho + \ln E_t\left[\frac{1+r_{t+1}}{(1-s_{t+1})Y_{t+1}/N_{t+1}}\right]$$

- Setting $s_t = s_{t+1} = \hat{s}$ and solving for $\hat{s}$, this implies:

$$\hat{s} = \alpha e^{n-\rho}$$

- confirms the conjecture that there is a solution with constant s.

Step 2: use optimal labor supply condition to find $\hat{\ell}$.

- Rewrite labor supply condition in terms of $\hat{s}$ using $c_t = (1-\hat{s})\frac{Y_t}{N_t}$, and take logs:

$$\ln\left[(1-\hat{s})\frac{Y_t}{N_t}\right] - \ln(1-\ell_t) = \ln w_t - \ln b$$

- Using $w_t = MPL = (1-\alpha)Y_t/(\ell_t N_t)$ and solving for ℓ, we get

$$\ell_t = \frac{1-\alpha}{(1-\alpha)+b(1-\hat{s})} = \hat{\ell}$$

- confirms that there is a solution with constant ℓ as well as constant s.

Implications for output dynamics

- From production function (taking logs):

$$\ln Y_t = \alpha \ln K_t + (1-\alpha)(\ln A_t + \ln L_t)$$

- Using definitions and assumptions on K, L, N and A, and the fact that $s = \hat{s}$ and $\ell = \hat{\ell}$, and after lots of boring (but easy) algebra, you get:

$$\tilde{Y}_t = (\alpha + \rho_A)\tilde{Y}_{t-1} - (\alpha\rho_A)\tilde{Y}_{t-2} + (1-\alpha)\epsilon_{A,t}$$

  - $\tilde{Y}_t$ = deviation of $\ln Y_t$ from its balanced growth-path.

- Bottom line: Transitory hump-shaped fluctuations around balanced growth path

- Positive (negative) productivity shock increases (decreases) output.

- fixed s and $\ell \rightarrow$ wages, consumption and investment are all highly pro-cyclical;

The general case

- Cannot be solved analytically;
- Log-linear approximation to the model
  - log-linearize around the steady-state
  - 1st order Taylor approximation in the logs of key variables
  - can be solved analytically
- Method of undetermined coefficients:
  1. conjecture the general functional form of the (log-linearized) solution;
  2. use the equations of the model to pin down coefficients' values.

Method of undetermined coefficients

- Define $\tilde{X}_t = \ln X_t - \ln X^*$
- Log-linearization around steady-state:

$$\tilde{C}_t \approx a_{CK}\tilde{K}_t + a_{CA}\tilde{A}_t + a_{CG}\tilde{G}_t$$

$$\tilde{L}_t \approx a_{LK}\tilde{K}_t + a_{LA}\tilde{A}_t + a_{LG}\tilde{G}_t$$

- Then use the two equilibrium conditions for household behavior to pin down the a's.

Optimal labor supply condition

- Labor-supply equilibrium condition:

$$\frac{c_t}{(1-\ell_t)} = \frac{w_t}{b}$$

- using the fact that $w = (1-\alpha)[K_t/(A_t L_t)]^{\alpha} A_t$ and taking logs

$$\ln c_t - \ln(1-\ell_t) = \ln\left(\frac{1-\alpha}{b}\right) + (1-\alpha)\ln A_t + \alpha \ln K_t - \alpha \ln L_t$$

- Take log-linear approximation around balanced growth path
  - it means that for both sides, we compute the (approximate) deviation from its steady state value.

$$\tilde{C}_t + \frac{\ell^{\star}}{1-\ell^{\star}}\tilde{L}_t = (1-\alpha)\tilde{A}_t + \alpha\tilde{K}_t - \alpha\tilde{L}_t$$

(see handout on Keats for the detail of this log-linearization)

- Log-linear approximation of the optimal ℓ condition

$$\tilde{C}_t + \frac{\ell^\star}{1-\ell^\star}\tilde{L}_t = (1-\alpha)\tilde{A}_t + \alpha\tilde{K}_t - \alpha\tilde{L}_t$$

- Substitute the general-form solutions for $\tilde{C}$ and $\tilde{L}$ into this:

$$a_{CK}\tilde{K}_t + a_{CA}\tilde{A}_t + a_{CG}\tilde{G}_t + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha\right)(a_{LK}\tilde{K}_t + a_{LA}\tilde{A}_t + a_{LG}\tilde{G}_t) =$$

$$= \alpha\tilde{K}_t + (1-\alpha)\tilde{A}_t$$

- ... and finally equate the coefficients on $\tilde{K}$, $\tilde{A}$ and $\tilde{G}$ on both sides

$$a_{CK} + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha\right)a_{LK} = \alpha; \quad a_{CA} + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha\right)a_{LA} = 1-\alpha$$

$$a_{CG} + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha\right)a_{LG} = 0$$

Coefficients restrictions from optimal labor supply condition:

$$a_{CK} + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha \right) a_{LK} = \alpha \tag{1}$$

$$a_{CA} + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha \right) a_{LA} = 1-\alpha \tag{2}$$

$$a_{CG} + \left(\frac{\ell^\star}{1-\ell^\star} + \alpha \right) a_{LG} = 0 \tag{3}$$

1. implies that either C or L (or both) must respond positively to $K \uparrow$;
2. implies that either C or L (or both) must respond positively to $A \uparrow$;
3. implies that the responses of C and L to $G \uparrow$ must have opposite sign;

Euler Equation

- Using the EE, we can add further restrictions.
- Pin down the a's as a function of the model parameters $(\alpha, g, n, \delta, \rho_A, \rho_G, \bar{G}, \rho, b)$;
- following D. Romer's wisdom, we skip this (very complicated) derivation;
- 'Impulse response functions': simulate effect of exogenous A and G shocks on the endogenous variables;
- If you feel going through derivation details would be helpful for you (eg, if you plan to work on macro modeling), you can study Campbell (1994) or similar;